Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/24/2017
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Particle Radiation-Induced Dysregulation of Protein Homeostasis in the Brain		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Menlo Park	**State:**	CA
Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	05/01/2014	**End Date:**	04/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Change in period of performance (original period of performance was 2/26/2014-2/25/2018) per PI and NSSC information (Ed., 12/9/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lin, Hua (SRI International) Shaler, Thomas (SRI International)		
Grant/Contract No.:	NNX14AE69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galacto-cosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. In normal biological systems, protein stability and system homeostasis are maintained through a tightly coupled post-translational modification process that is catalyzed by the ubiquitin-proteasome system for protein turnover. This process is necessary for modifying the correct protein in the correct cellular location at the correct time to control important processes, e.g., cell-cycle regulation, the maintenance of genomic stability, removal of damaged or misfolded proteins, autophagy, apoptosis, and gene transcription. Depending on the radiation dose and beam quality, particle radiation may cause significant damage to proteins and oxidative stress in the cells in the central nervous system, leading to dysfunction of the cellular ubiquitin-related pathways, an increase in the levels of extracellular ubiquitin, and persistent subsequent downstream effects on regulation of stress-associated signaling pathways in many different cell types within the central nervous system (CNS), precipitating neurodegenerative disease. We hypothesize that quantitative changes in cellular ubiquitin pools occur when neuronal cells are exposed to particle radiation and aim to study the impact of protein degradation after particle radiation using both in vitro and in vivo model systems. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect that results associated with the regulation of protein processing and degradation in brain tissues obtained from our studies will be useful to understand molecular mechanisms of action in normal neuronal cell responses in particle radiation clinical therapy scenarios.
Task Progress:	Our strategy is to use an in vitro/in vivo approach to assess changes in primary mouse and human neuronal cells and to compare these changes with results from the animal model. Three primary mouse neuronal stem cells, representing cells from the striatum, cortex, and ventral mesencephalon of the brain and human neural stem cells were used in these studies. C57Bl6 mice were used as the animal model. Animals and cells were irradiated with either 250 MeV proton or 1 GeV/n Fe ions. Cells lysates were harvested 24 hr post irradiation and brain tissues from various regions of the brain of the irradiated and sham –treated animals were harvested at 1, 8, or 11 weeks post irradiation. Samples were analyzed with a method that provides both quantitative levels of ubiquitin pools and relative levels for global proteome changes using capillary HPLC coupled with high-resolution mass spectrometry. There are 7 lysine (or K-) residues in the ubiquitin molecule, each serving as points of ubiquitination. K-48 chains were first identified and best characterized in mechanisms of proteolysis. K-63 linkage serves as a molecular platform for protein interactions in DNA damage repair and protein trafficking. K-11 linkage is less well known but is associated with proteosomal degradation. We observed that particle radiation quality plays an important role in the level and linkage of protein ubiquitination in neuronal cells and brain tissues. In particular, results from our analysis show that K-48 linkages are higher in the Fe-irradiated hippocampus tissue while similar changes in the Ub linkage pattern was noted in the affected brain regions in post-mortem tissue from Alzheimer patients. On the other hand, there was no significant difference was found in any of the 3 lysine-linked pUb chains in hippocampal tissues harvested from animals up to 8 weeks after proton exposures. We found increased linkage-dependent pUb in human neuronal cells after particle radiation exposure. On one hand, K-63 linkage was significantly higher in cells that were exposed to Iron ions relative to the controls. On the other hand, K-48 linkage was significantly higher in cells that were exposed to protons relative to the controls, while no significant changes were noted in the K-63 linkages. These results suggest that particle radiation with different linear energy transfers (LETs) may trigger alternate pathways for protein degradation in human cells. Representative cell types from three different regions of the mouse brain were used in our studies. Changes in pUb linkages were significant in specific cell lineages suggesting that specialized regions in the brain may be more susceptible to LET-dependent protein degradation than others. In conclusion, particle fluences play a role in targeted protein degradation in neuronal tissues. Our results suggest that long-term changes in protein turnover pathways related to the ubiquitin proteasome system can occur in brain regions after particle irradiation.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Abstracts for Journals and Proceedings	Chang PY. "Overview of Particle Effects on CNS." Presented at the 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Chang PY, Shaler TA, Lin H. "Particle-radiation induced dysregulation of protein homeostasis in the brain." Presented at the 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Abstracts for Journals and Proceedings	Chang PY. "Translating Discovered Drug Candidates from the Benchtop to Human Use, an Overview of Product Development Process." Presented at the 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017